Documentation for Manual Image Captioning of Tomato Leaf Images

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1. Introduction

This document describes the manual image captioning work carried out for tomato leaf images. The task involved observing each image and writing multiple descriptive captions based on the visible features of the leaf. Instead of assigning only a disease label, every image was annotated with five unique captions that describe characteristics such as color, spots, discoloration, vein patterns, and leaf shape.

The captions were written manually after carefully examining each image to ensure that every observation was supported by the image. Once the initial annotations were completed, the captions were reviewed by the mentor. The review helped identify incorrect or incomplete observations, which were then corrected to produce the final verified set of captions.

2. Objective

The primary objective of this task was to develop a high-quality caption dataset for tomato leaf disease images by manually creating multiple natural language descriptions for every image.

The specific objectives of this work were:

- To observe every tomato leaf image manually instead of relying only on disease labels.
- To write five unique captions for every image.
- To ensure that each caption focused on different visual characteristics of the leaf.
- To avoid repetitive sentence structures commonly generated by AI systems.
- To write captions using simple and natural English so that they resemble observations made by a human.
- To avoid making assumptions or mentioning details that were not visible in the image.

3. Dataset Overview

The dataset used in this work consists of tomato leaf images collected from six different disease categories. Each disease category contains five representative images. Every image was manually analyzed and annotated with five unique captions.

Disease Categories

- Bacterial Spot
- Early Blight
- Leaf Mold
- Mosaic Virus
- Target Spot
- Yellow Leaf Curl Virus

Dataset Statistics

Parameter	**Value**
Disease Categories	6
Images per Category	5
Total Images	30
Captions per Image	5
Total Captions	150

Instead of generating captions automatically, each caption was written manually after visually examining the image. This ensured better accuracy and diversity.

4. Annotation Methodology

The annotation process followed a structured workflow to ensure consistency across all tomato leaf images. Each image was manually examined, and five descriptive captions were written based on the visible characteristics of the leaf. The workflow followed during annotation is described below.

Step 1 – Image Observation

Each image was carefully examined before writing any captions. The focus was on observing only the features that were clearly visible in the image rather than relying on the disease category or making assumptions about the condition of the leaf.

During this stage, the following characteristics were observed:

- Overall leaf appearance
- Color variation across the leaf
- Brown or dark patches

- Yellowing or discoloration
- Distribution of spots or damaged areas
- Leaf margins and edges
- Shape of the leaf
- Vein visibility
- Curling or folding of the leaf
- Relative position of visible features

Only observations that could be confirmed visually were included in the captions.

Step 2 – Caption Generation

After observing the image, five unique captions were written for each leaf image. Rather than rephrasing the same sentence multiple times, each caption highlighted a different visual aspect of the leaf to provide a more complete description.

The captions collectively described characteristics such as the overall appearance of the leaf, the distribution of discoloration, the position and size of spots, vein visibility, leaf margins, and other noticeable visual patterns.

Different sentence structures and descriptions were used to ensure that each caption contributed new information while maintaining consistency with the image.

Step 3 – Review and Refinement

Once the initial captions were completed, they were reviewed against the corresponding images to verify that every observation was supported by the visual content.

The reviewed captions were then validated by the mentor. Based on the feedback received, unsupported observations were removed, missing visual details were added, and the wording of several captions was refined to improve accuracy, clarity, and consistency. The final dataset consists of mentor-verified captions for all tomato leaf images.

5. Caption Writing Guidelines

The following guidelines were followed while writing the captions to maintain consistency throughout the dataset.

5.1 Describe Only What Is Visible

Every caption was written based only on features that could be directly observed in the image. Statements that required assumptions or additional context were avoided.

Example

Several brown patches are scattered across the leaf.

The leaf is infected with a fungal disease.

5.2 Focus on Visual Features

The captions describe observable characteristics such as color variation, spots, discoloration, veins, leaf margins, and overall appearance. Features that were not clearly visible were not included.

5.3 Avoid Disease Identification

Although the images belonged to different disease categories, the captions were written to describe the leaf rather than identify or explain the disease.

Example

Yellowing is visible along the margins of the leaf.

The leaf is affected by Yellow Leaf Curl Virus.

5.4 Ensure Caption Diversity

Five captions were written for every image. Each caption focused on a different observation so that the complete set provided a comprehensive description without repeating the same information.

5.5 Maintain Consistency

The captions were written in a consistent style across all disease categories. Similar observable features were described using consistent terminology while ensuring that each caption remained unique to the corresponding image.

6. Quality Assurance

After the initial set of captions was completed, all annotations were reviewed internally to check for consistency and accuracy before submitting them for mentor review.

During this review, the following aspects were verified:

- Accuracy of visual observations
- Grammar and sentence structure
- Readability
- Consistency in writing style
- Diversity across the five captions for each image

Captions containing repetitive descriptions were revised to focus on different visual aspects of the image. Any unsupported observations or unnecessary wording identified during the review were corrected to ensure that the captions remained based only on the visible content of the image.

7. Mentor Review

After completing the manual annotation process, the captions were submitted to the mentor for review.

Each caption was verified against its corresponding image to ensure that the descriptions accurately represented the visible features. The review mainly focused on:

- Correctness of observations
- Addition of missing visual details
- Removal of unsupported statements
- Sentence clarity and grammar
- Consistency across captions

The mentor provided feedback by marking incorrect or unnecessary text in **red** and adding suggested modifications in **green**. These changes were carefully incorporated into the final version of the captions. The review process improved the overall quality, accuracy, and consistency of the dataset.

8. Challenges Faced

Although the dataset consisted of only thirty images, writing five unique captions for each image required careful observation and planning.

One of the main challenges was ensuring that all five captions described different aspects of the same image without becoming repetitive. Instead of rewriting the same sentence with different words, each caption was written to highlight a different visible feature such as color variation, spot distribution, vein visibility, leaf margins, or overall appearance.

Another challenge was ensuring that every statement was supported by the image. During the mentor review, a few captions contained observations that were considered too subjective or not clearly visible. These were revised to make the descriptions more objective and image-grounded.

Maintaining consistency in writing style across all disease categories while keeping the captions natural and descriptive was another important aspect of the annotation process.

9. Final Dataset Summary

The completed annotation task resulted in:

- **30** tomato leaf images
- **6** disease categories
- **5** captions for each image
- **150** manually written captions

The captions were reviewed and refined based on mentor feedback to improve their accuracy and consistency. The final dataset consists of descriptive captions that are directly based on the visible characteristics of each tomato leaf image.

10. Conclusion

The manual image captioning task was successfully completed for all tomato leaf images. Each image was carefully examined and annotated with five unique captions describing its visible characteristics. The captions were written from different perspectives to provide a comprehensive description of the image while avoiding repetitive observations.

Following the mentor review, the captions were refined by correcting unsupported statements, incorporating additional visible details, and improving sentence clarity. The final set of captions provides accurate and consistent descriptions that can be used for image captioning, image-text understanding, and future vision-language model development.